Logistic Regression and Decision Boundaries

Logistic Regression predicts binary probabilities p in [0, 1] using the Sigmoid activation function: sigma(z) = 1 / (1 + exp(-z)), where z = theta^T * x.

Binary Cross-Entropy (Log-Loss): J(theta) = - (1/m) * sum(y_i * log(p_i) + (1 - y_i) * log(1 - p_i)). The decision boundary is the hyperplane theta^T * x = 0 separating predicted classes where probability equals 0.5.

Confusion Matrix, Precision, Recall, and F1-Score

Evaluation of classification models requires parsing the Confusion Matrix:
- True Positives (TP): Positive cases correctly predicted positive.
- False Positives (FP - Type I error): Negative cases incorrectly predicted positive.
- False Negatives (FN - Type II error): Positive cases incorrectly predicted negative.
- True Negatives (TN): Negative cases correctly predicted negative.

Metrics:
- Precision = TP / (TP + FP): Proportion of predicted positives that are actual positives.
- Recall (Sensitivity) = TP / (TP + FN): Proportion of actual positives captured by the model.
- F1-Score = 2 * (Precision * Recall) / (Precision + Recall): Harmonic mean balancing precision and recall.

ROC Curve, AUC, and Imbalanced Classification

The Receiver Operating Characteristic (ROC) curve plots True Positive Rate (Recall) against False Positive Rate (FPR = FP / (FP + TN)) across varying probability thresholds from 0 to 1.

Area Under the ROC Curve (AUC) measures the model's ability to rank positive instances higher than negative instances. An AUC of 0.5 represents random guessing, while 1.0 represents perfect discrimination. For heavily imbalanced datasets (e.g., fraud detection), Precision-Recall curves provide a much more informative metric than ROC-AUC.